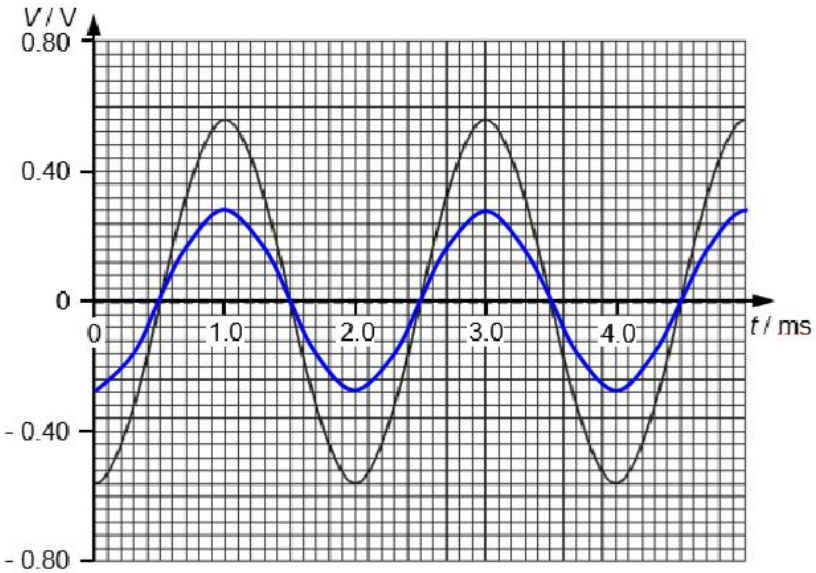
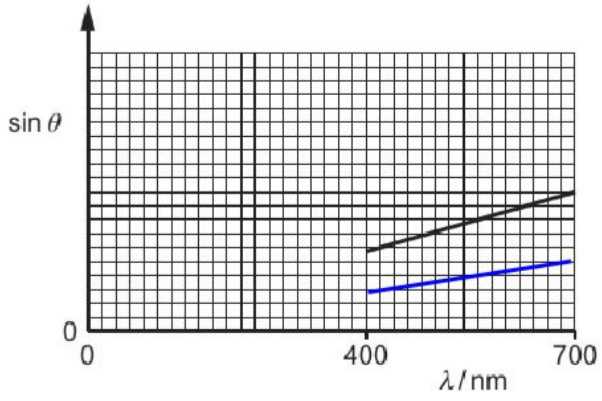
8	(a)(i)	$T = 4 \times 0.50 \times 10^{-3}$ $T = 2.0 \times 10^{-3}$ s $f = \dfrac{1}{2.0 \times 10^{-3}}$ $f = 500$ Hz	A1
	(a)(ii)	 Intensity of the wave is proportional to the square of its amplitude. Since the intensity is reduced to one quarter, the amplitude is reduced to half. The period of the wave remains unchanged.	B1 B1
	(b)(i)	The incident sound wave (travels towards the metal sheet and) reflects at the metal sheet, creating a reflected wave that travels in opposite direction. Since both waves possess the same frequency and amplitude, they continuously superpose as they pass through each other, creating a stationary wave.	B1 B1 A0
	(b)(ii)	$d = 1.5\lambda$ $1.05 = 1.5\lambda$ $\lambda = 0.70$ m $v = f\lambda$ $v = 500 \times 0.70$ $v = 350$ m s^{-1}	C1 C1 A1
	(b)(iii)	Since the material absorb sound, the reflected wave has a smaller amplitude than the incident wave. Hence there is incomplete cancellation at the nodes.	M1

		The amplitude registered at the minimum position increases. BOD if students state that the amplitude is "no longer zero"	**A1**
	(c)(i)1.	Since point Q is at the first order maximum. phase difference = $1 \times 2\pi = 360°$ (accept 0°)	 **A1**
	(c)(i)2.	Since point P is at the second order minimum, path difference = $1.5\lambda = 1.5 \times 610$ $= 915 \approx 920$ nm	 **A1**
	(c)(ii)	$x = \frac{1}{4} \times 22 \times 10^{-3} = 5.5 \times 10^{-3}$ $\frac{ax}{D} = \lambda$ $\frac{a \times 5.5 \times 10^{-3}}{2.7} = 610 \times 10^{-9}$ $a = 2.995 \times 10^{-4} \approx 3.0 \times 10^{-4}$ m	**C1** **A1**
	(c)(iii)	All component wavelengths of white light have a path difference of zero at the center, causing them to superpose completely constructively. The central bright fringe appears as a (sharp) white line. Since fringe separation is directly proportional to the wavelength, The side fringes on either side of this center fringe appear as continuous colored spectra where violet is closest to the center and red is furthest away.	 **B1** **B1**
	(d)(i)	$G = \frac{\sin\theta}{\lambda}$ $a\sin\theta = n\lambda$ $d = n\frac{\lambda}{\sin\theta}$ $d = \frac{4}{G}$	**C1** **A1**
	(d)(ii)		

		$a \sin\theta = n\lambda$ $\frac{\sin\theta}{\lambda} = \frac{n}{a}$ Since the order of the maximum is reduced from 4 to 2, the gradient of the line should be halved.	
		straight line from 400 nm to 700 nm that is always below the printed line	**M1**
		straight line has half the gradient as the printed line.	**A1**